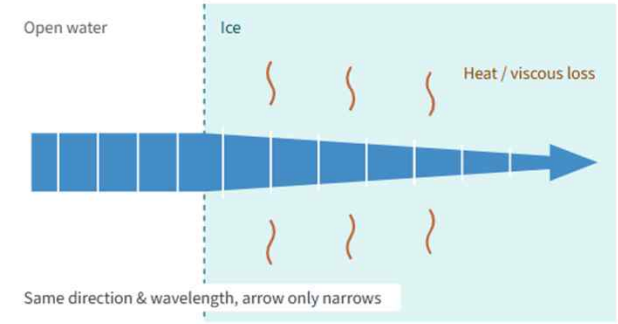
Review of paper Bai et al. 2020

Donghyun Kim

Intro

Attenuation (dissipation)

Energy is lost



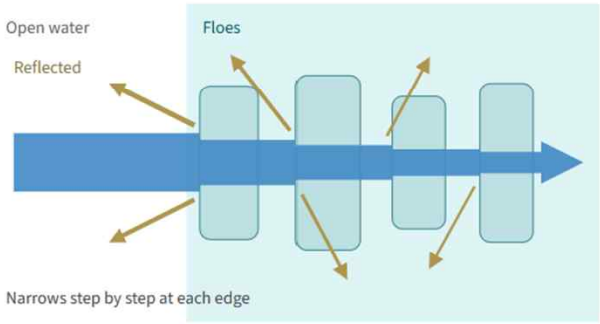
Transmitted | Dissipated as heat (gone)

Fate of 100 units of incident energy (illustrative)

As the ice flexes and friction and viscosity act between and beneath floes, wave energy is **converted to heat and lost**. It decays continuously and exponentially with distance into the ice.

Scattering

Energy is redistributed (conserved)



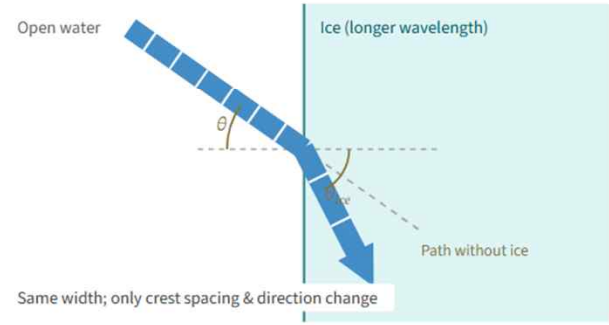
Transmitted | Reflected / scattered (still exists)

Fate of 100 units of incident energy (illustrative)

Each time the wave meets a **floe edge**, part of it is reflected back or spread sideways. No energy is lost, but the forward share shrinks, so deep in the ice it looks like attenuation.

Refraction

Energy unchanged, direction bends



Transmitted 100 (in a new direction)

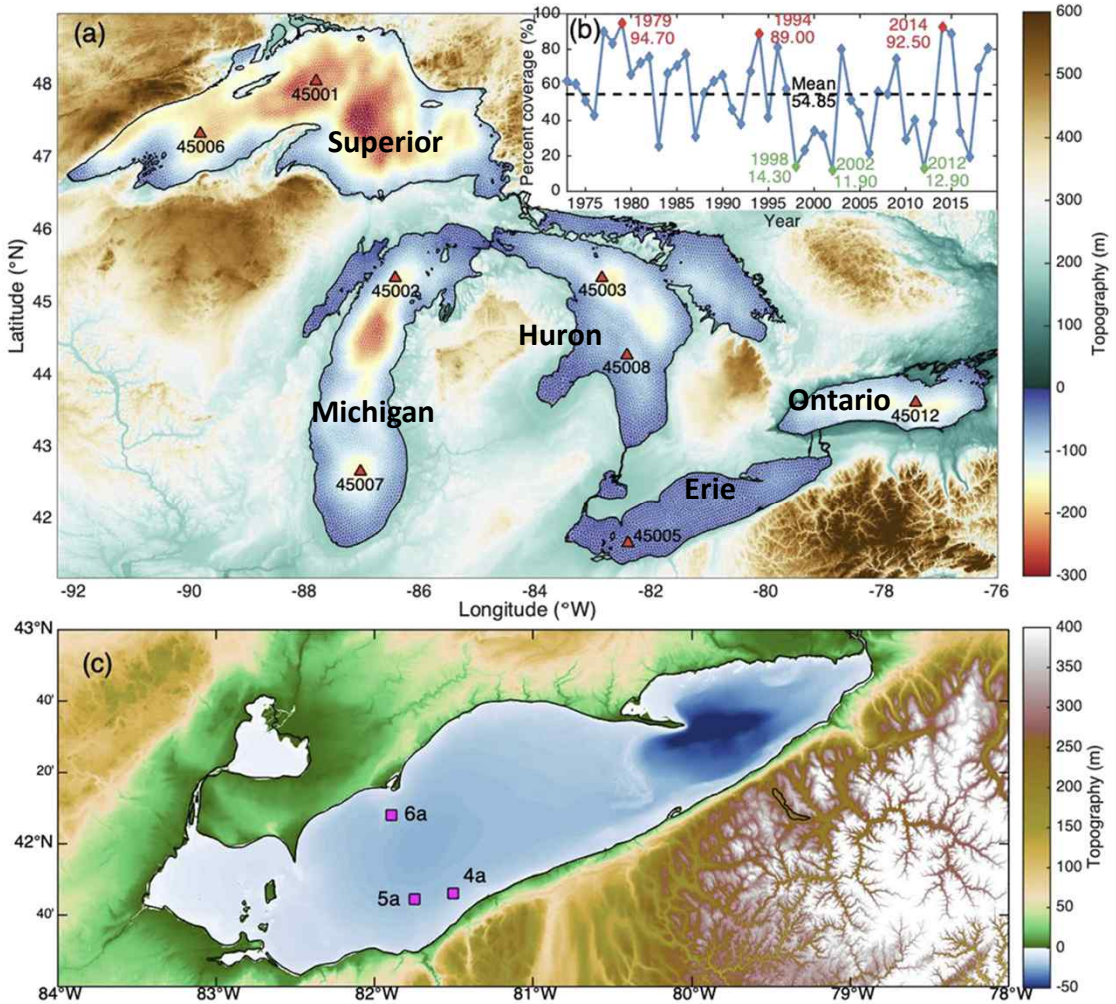
Fate of 100 units of incident energy (illustrative)

Because of its flexural rigidity, the ice sheet gives a **different wavelength and speed** at the same period. A wave arriving obliquely bends at the boundary, like light entering water. Energy is not reduced.

Q.o.I : How much does ice-induced attenuation affect wave propagation?

What observation data is used for validation? ※ Section : Intro

Obs. source	Timeline	What is validated	Outputs of source
NDBC buoy (8 spots)	2014 May – Nov	SWAVE model (w/o ice) NARR model	SWH(= H_s), U_10
AWAC (Lake Erie 3 spots)	2011 Jan – Mar	CICE-SWAVE model	SWH(= H_s)
NIC satelite	Daily during ice seasons	CICE model(C_ice)	C_ice



Why choose this timeline?

1. Choose year characterized by high sea ice concentration
2. Availability of winter obs.

FVCOM · UG-CICE · SWAVE

Information exchanged
Not considered in Bai

"Partly coupled wave-ice model": all three modules run together inside FVCOM on the same unstructured grid (~3.5 km, 21 σ-layers) with the same time step (60 s).

ATMOSPHERIC FORCING (EXTERNAL INPUT)

NARR reanalysis

3-hourly, 32 km horizontal resolution. Drives all three modules: 10 m wind U_{10}, air temperature, humidity, radiation, pressure, etc. Bai identifies NARR winds being 0.3–0.9 m/s weaker on average than NDBC buoy observations as one source of SWH error.

Wind stress, heat & moisture fluxes

Air temp, wind, radiation, humidity

U_{10} → wind input S_{in}

LAKE CIRCULATION MODEL

FVCOM

3D primitive equations (momentum, continuity, heat, salt). Great Lakes-FVCOM (Bai et al. 2013). Internal mode 60 s, split ratio 20, leapfrog time stepping, minimum depth 10 m.

Input
NARR: wind stress, heat flux
UG-CICE: ice–water stress, heat & freshwater fluxes*

Output
Currents u, v · water level ζ · water temp T_w · salinity
→ UG-CICE (water temp, currents)*
→ SWAVE ✕ (not passed)
※ Normally currents (U = (u,v)) and water level ζ enter SWAVE (LHS), but to isolate the ice attenuation effect Bai sets U = 0 and keeps C_{theta}, C_{sigma} (depth = base depth + water level ζ) fixed.

Water temp, currents*
(freeze/melt, drift)

Two-way
internal coupling

Ice–water stress,
heat & freshwater fluxes*

ICE MODEL

UG-CICE (= CICE)

Unstructured-grid finite-volume version of CICE (Los Alamos) (Gao et al. 2011). Ice thermodynamics/dynamics model **internally coupled** to FVCOM.

Input
NARR: air temp, wind, radiation
FVCOM: **water temp, currents***
SWAVE: ✕ (no wave feedback)

Output
Ice concentration C_{ice} · thickness H_i · ice velocity
→ SWAVE (C_{ice}, H_i)
→ FVCOM (stress, fluxes)*

C_{ice}, H_i
passed every 60 s

One-way
ice → wave

Wave → ice
feedback not modeled
(no ice fracture)

SPECTRAL WAVE MODEL

FVCOM-SWAVE (= SWAVE)

SWAN-based third-generation model (Qi et al. 2009). 40 frequencies (0.04–1 Hz) × 36 directions. Komen growth/whitecapping, Madsen bottom friction, Hasselmann quadruplets + **IC4 ice attenuation**.

Input
NARR: U_{10}
UG-CICE: C_{ice} → blocking $(1-C_{ice})$, H_i → α (IC4M3·M7)

WAVE ACTION BALANCE

$$\partial N/\partial t + \nabla\cdot[(C_g+U)N] + \partial(C_\sigma N)/\partial\sigma + \partial(C_\theta N)/\partial\theta = S_{tot}/\sigma$$

$$S_{tot} = (1-C_{ice})(S_{in}+S_{ds,w}+S_{ds,br}) + S_{nl} + S_{ds,b} + C_{ice}S_{ice}$$

Output
Action density spectrum $N(\sigma,\theta)$ · **significant wave height SWH · period T** · direction
→ not passed to other modules ✕ (final product)

✕ Wave–current interaction not considered — no currents/water level → SWAVE, no radiation stress/wave mixing → FVCOM

VALIDATION DATA · CIRCULATION

Currents and water temperature are not separately validated in Bai 2020 (uses Bai et al. 2013 setup)

VALIDATION DATA · ICE

NIC satellite ice concentration — compared with Feb/Mar 2014 monthly-mean C_{ice} spatial patterns

VALIDATION DATA · WAVES

8 NDBC buoys SWH (2014, ice-free season) · **3 AWAC** sites Lake Erie SWH (Feb 2011, ice season)

NARR → all modules
One-way forcing

FVCOM ⇄ UG-CICE
Two-way (ice model internally coupled in FVCOM)

UG-CICE → SWAVE
One-way: C_{ice} (blocking), H_i (attenuation α) — the paper's key new feature

SWAVE → UG-CICE ✕
No wave-induced ice fracture/melt → added later in Javaherian 2025

FVCOM ⇄ SWAVE ✕
Wave–current interaction disabled (U term unused)

CICE · Icepack · WIFF — WAVEWATCH III

Horvat & Roach (2022), WIFF 1.0. WIFF is not a separate model but **a subroutine inside CICE (Icepack)** — called and returned for every grid cell at every time step.

Data passed (evaluation run, 2005)
Only in the training-data run (2009)
Function call / return (same program)

PRECOMPUTED FILE (OFFLINE)

Output of the Roach et al. (2019) coupled CICE–WW3 run

This paper does not run the wave model; it only reads the output file

Read spectrum file

ATMOSPHERIC FORCING (EXTERNAL INPUT)

JRA55 reanalysis — 3-hourly

Air temperature, wind, humidity, long/shortwave radiation, precipitation → drive sea-ice thermodynamics (freezing/melting) and dynamics (drift/deformation) → **concentration C and thickness H** updated every time step

Heat & momentum fluxes

SPECTRAL WAVE MODEL

WAVEWATCH III

In the 2005 evaluation run, **precomputed time-varying spectra are used as forcing** (not included in run time).

Input (2009 coupled run only)
CICE: C, H, mean floe size → wave attenuation in ice

Output
$S(\lambda_i)d\lambda_i = \int S(\lambda_i,\Theta)d\Theta$ (Eq. 1) → **25 wavelength bins**

$S(\lambda)$ **25 bins**
6-hourly

C, H, size
(2009 only)

GLOBAL SEA-ICE MODEL (STAND-ALONE, NOMINAL 1°, 384×320)

CICE6.1.4 + Icepack

Thermodynamics: freezing/melting change H · Dynamics: drift/convergence change C
FSD 12 categories × thickness categories (Roach et al. 2019)

Grid-cell state — concentration C · mean thickness H · floe size distribution $f(r)$ · received spectrum $S(\lambda)$ — Updated every time step

Read state

WIFF pre-filter — called only if $H < 10$ m, $C > 0.01$, $H_s = 4\sqrt{E} > 0.1$ m

27 call arguments
$[S(\lambda_1) \dots S(\lambda_{25}), H, C]$

SUBROUTINE **WIFF 1.0 — runs as either SP or NN**

NN ① CLASSIFIER
Input 27 → 100 → 100 → softmax 2
Output fracture score $\chi \in [0, 1]$

$\chi < \chi_{crit} = 0.54$?
Yes → fracture, run ②
No → return (no change)

NN ② HISTOGRAM NETWORK
Input same 27 → 100×5 → softmax 12
Output $\hat{\Lambda}_i dr_i$, 12 bins, sum = 1

Return $\Lambda_i dr_i$

SP-WIFF (ORIGINAL, STOCHASTIC) — SAME 27 INPUTS
S1 spectrum → 10 km 1-D strain field $\eta(x)$, random phases
S2 distances between failing strain extrema → fracture-length histogram $\Lambda(r)$
S3 iterate until $\Delta\Lambda < 5\times10^{-4}$ (SP-WIFF1 = one pass)
→ same output $\Lambda_i dr_i$. Used to generate the NN **ground truth (training labels)**

TRAINING (OFFLINE, ONCE)
5.1 M (27 inputs → SP-WIFF output) pairs collected from the 2009 run · Keras · 70/30 split
Loss RSE = $\Sigma|\hat{\Lambda}_i - \Lambda_i|r_i dr_i \cdot \chi_{crit}$ chosen to minimize validation error rate

JRA55 → CICE
One-way atmospheric forcing. Through the thermodynamic/dynamic equations it becomes C, H → 2 of the 27 WIFF inputs

WW3 → CICE
Evaluation run (2005): one-way offline file. Training-data run (2009): two-way coupled → 25 of the 27 WIFF inputs

CICE → WIFF → CICE
Not model coupling but a **function call**: pass 27, receive 12. Different from Bai's FVCOM ⇄ UG-CICE (communication between two models)

No observational validation
Ground truth = SP-WIFF output. Only how well NN-WIFF reproduces SP-WIFF is evaluated

Model configuration - Equation in Wave Model(SWAVE) ※ Section 2.1-2.2

SWAN / SWAVE(original)

$$\frac{\partial N}{\partial t} + \nabla \cdot [(\vec{c}_g + \vec{U})N] + \frac{\partial C_\sigma N}{\partial \sigma} + \frac{\partial C_\theta N}{\partial \theta} = \frac{S_{in} + S_{nl} + S_{ds}}{\sigma}$$

LHS: Energy Transport (where the energy goes) / RHS : Energy generated and lost

SWAVE (Bai's version)

$$\frac{\partial N}{\partial t} + \nabla \cdot \left[\left(\vec{\mathbf{c}_g} + \vec{\mathbf{U}} \right) N \right] + \frac{\partial C_\sigma N}{\partial \sigma} + \frac{\partial C_\theta N}{\partial \theta} = \frac{(1 - C_{\text{ice}}) \left(S_{\text{in}} + S_{\text{ds,w}} + S_{\text{ds,br}} \right) + S_{\text{nl}} + S_{\text{ds,b}} + S_{\text{ice}}}{\sigma},$$

S_in : wind input
S_nl : nonlinear wave-wave interaction
S_ds: dissipation [S_ds,w(whitecapping), S_ds,br(depth induced wave breaking), S_ds,b(bottom friction)]
S_ice: wave energy sink due to ice damping
C_ice: ice concentration
1 – C_ice : "blocking effect"(ice inhibits wind input)

SWAVE

$$\frac{\partial N}{\partial t} + \nabla \cdot \left[\left(\vec{\mathbf{c}_g} + \vec{\mathbf{U}} \right) N \right] + \frac{\partial C_\sigma N}{\partial \sigma} + \frac{\partial C_\theta N}{\partial \theta} = \frac{(1 - C_{\text{ice}}) \left(S_{\text{in}} + S_{\text{ds,w}} + S_{\text{ds,br}} \right) + S_{\text{nl}} + S_{\text{ds,b}} + S_{\text{ice}}}{\sigma},$$

Scattering kernel

Attenuation

$$S_{\text{ice}} = -c_g E \left(\alpha_{\text{scat}} + \alpha_{\text{dis}} \right) + c_g \int_0^{2\pi} EK(\theta - \theta') \, \mathrm{d}\theta'$$

Group velocity　Scattering　Dissipation　Energy density

$$S_{ice} = -\alpha E C_g$$

- E = wave spectral density
  (Current wave energy)

- C_g = wave group velocity
  (How fast wave energy is moving)

- α = attenuation coefficient

α ↑/ ↓ → attenuation ↑ / ↓

※ Bai (2020), Javaherian (2025)

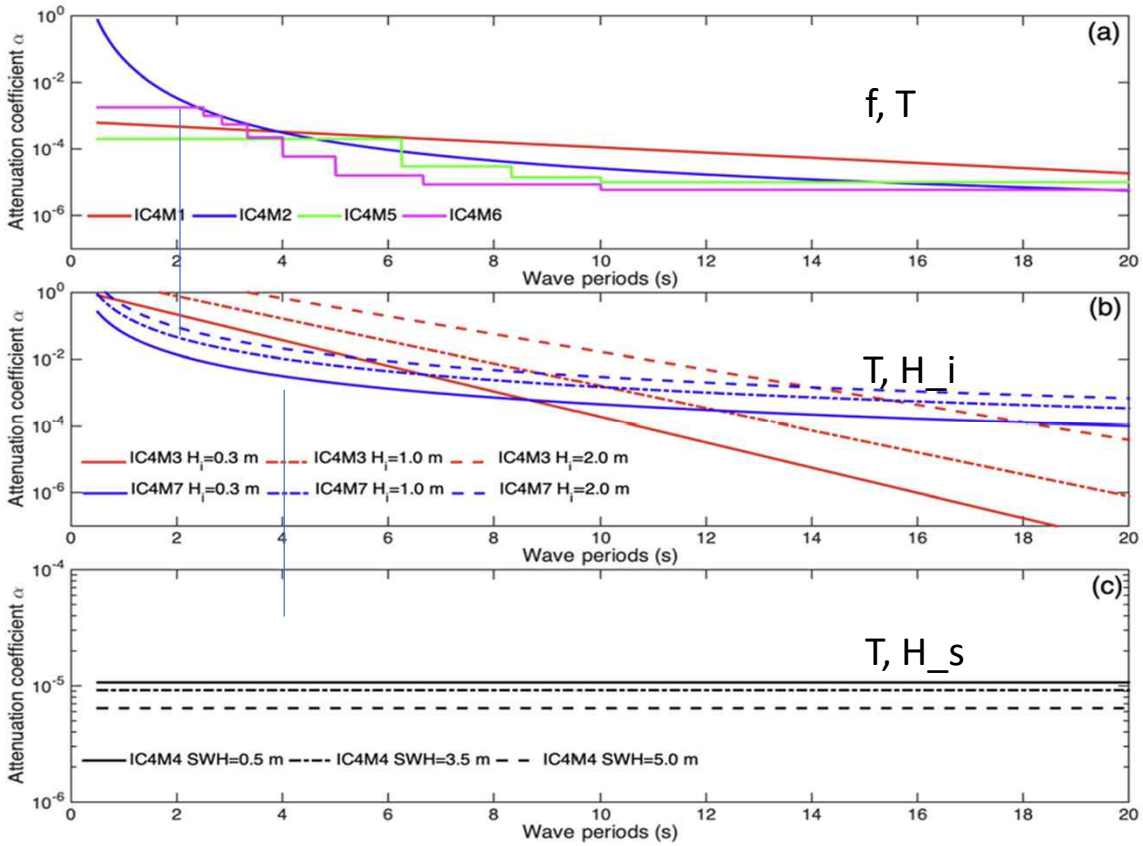
Apply WavewatchⅢ model's IC4 formula to calculate α

Table 1 Summary of the origins and formulas in IC4, where α is attenuation coefficient, T denotes wave period, H_i is ice thickness, SWH is significant wave height, and f is wave frequency

Method	Reference	Formula
IC4M1	Wadhams et al. (1988)	$\alpha = e^{(-0.18 * T - 0.73)}$
IC4M2	Meylan et al. (2014)	$\alpha = \frac{2.12*10^{-3}}{T^2} + \frac{4.59*10^{-2}}{T^4}$
IC4M3	Kohout and Meylan (2008) Horvat and Tziperman (2015)	$\alpha = e^{\left(-0.3203 + 2.058 H_i - 0.9375 T - 0.4269 H_i^2 + 0.1566 H_i T + 0.0006 T^2\right)}$
IC4M4	Kohout et al. (2014)	$\alpha = \begin{cases} 2*5.35*10^{-6} & (\text{SWH} \le 3) \\ \frac{2*16.05*10^{-6}}{\text{SWH}} & (\text{SWH} > 3) \end{cases}$
IC4M5	Doble et al. (2015)	$\alpha = \begin{cases} 2*5.0*10^{-6} & (f < 0.1) \\ 2*7.0*10^{-6} & (0.1 \le f < 0.12) \\ 2*1.5*10^{-5} & (0.12 \le f < 0.16) \\ 2*1.0*10^{-4} & (f \ge 0.16) \end{cases}$
IC4M6 (IC4M6H)	Rogers et al. (2016)	$\alpha = \begin{cases} 2*2.94*10^{-6} & (f < 0.1) \\ 2*4.27*10^{-6} & (0.1 \le f < 0.15) \\ 2*7.95*10^{-6} & (0.15 \le f < 0.20) \\ 2*2.95*10^{-5} & (0.20 \le f < 0.25) \\ 2*1.12*10^{-4} & (0.25 \le f < 0.30) \\ 2*2.74*10^{-4} & (0.30 \le f < 0.35) \\ 2*4.95*10^{-4} & (0.35 \le f < 0.40) \\ 2*8.94*10^{-4} & (f \ge 0.40) \end{cases}$
IC4M7	Doble et al. (2015)	$\alpha = 0.2 * T^{-2.13} * H_i$

Numerical Experiment - Comparing Attenuation Coefficient formulas ※ Section 2.2

Method	Formula
IC4M1	$\alpha = e^{(-0.18 * T - 0.73)}$
IC4M2	$\alpha = \frac{2.12*10^{-3}}{T^2} + \frac{4.59*10^{-2}}{T^4}$
IC4M3	$\alpha = e^{(-0.3203+2.058H_i-0.9375T-0.4269H_i^2+0.1566H_iT+0.0006T^2)}$
IC4M4	$\alpha = \begin{cases} 2*5.35*10^{-6} & (\mathrm{SWH} \le 3) \\ \frac{2*16.05*10^{-6}}{\mathrm{SWH}} & (\mathrm{SWH} > 3) \end{cases}$
IC4M5	$\alpha = \begin{cases} 2*5.0*10^{-6} & (f < 0.1) \\ 2*7.0*10^{-6} & (0.1 \le f < 0.12) \\ 2*1.5*10^{-5} & (0.12 \le f < 0.16) \\ 2*1.0*10^{-4} & (f \ge 0.16) \end{cases}$
IC4M6 (IC4M6H)	$\alpha = \begin{cases} 2*2.94*10^{-6} & (f < 0.1) \\ 2*4.27*10^{-6} & (0.1 \le f < 0.15) \\ 2*7.95*10^{-6} & (0.15 \le f < 0.20) \\ 2*2.95*10^{-5} & (0.20 \le f < 0.25) \\ 2*1.12*10^{-4} & (0.25 \le f < 0.30) \\ 2*2.74*10^{-4} & (0.30 \le f < 0.35) \\ 2*4.95*10^{-4} & (0.35 \le f < 0.40) \\ 2*8.94*10^{-4} & (f \ge 0.40) \end{cases}$
IC4M7	$\alpha = 0.2 * T^{-2.13} * H_i$



Results

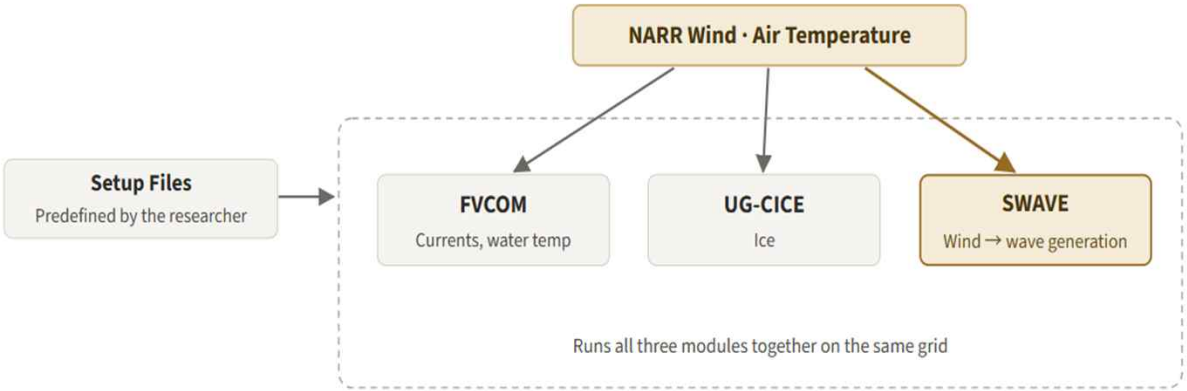
1. High-frequency waves gets more attenuation (f↑ ⇒ T↓ ⇒ α↑)
2. Thick ice gets more attenuation (H_i↑⇒α↑)
3. High waves relatively has less energy loss(H_s↑ ⇒α ↓)

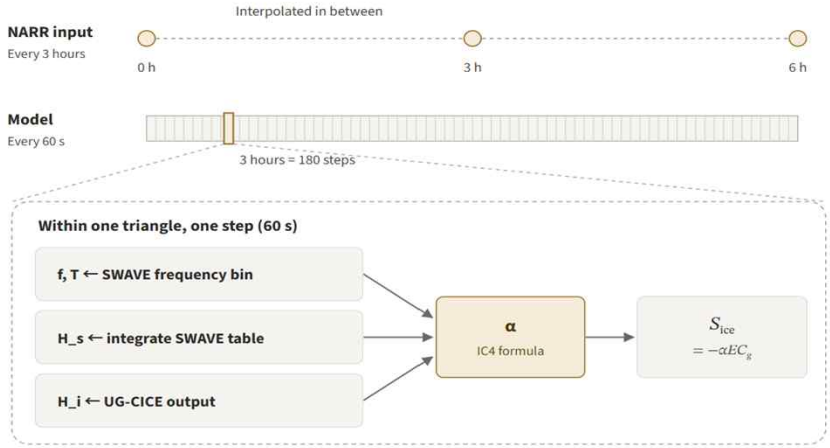
For the same T and conditions, α can differ by several orders of magnitude depending on the scheme.

Numerical Experiments – Settings

※ Section 2.3

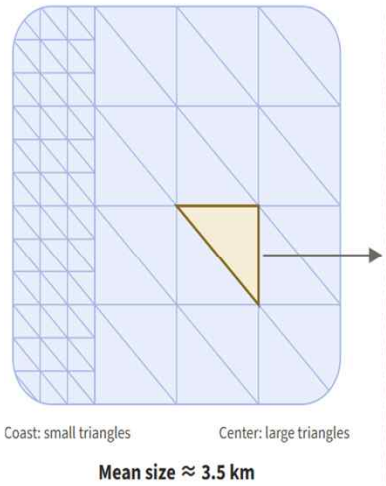
Q : What determines α in the Great Lakes?

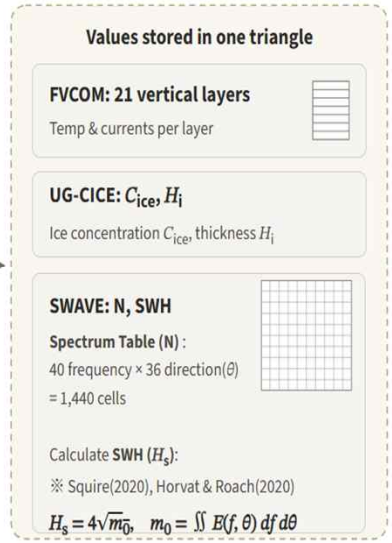


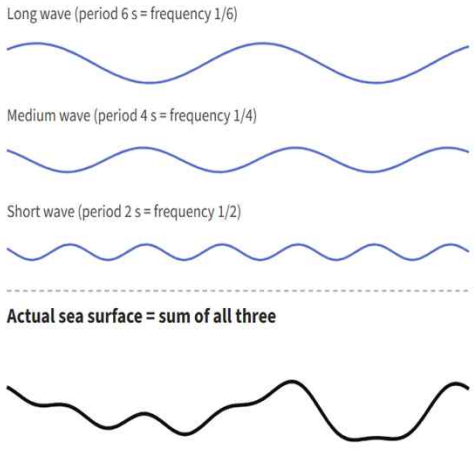


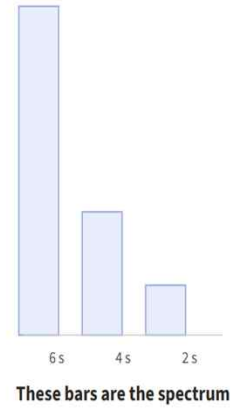
SETUP

SET	explanation
Horizontal grid	Unstructured triangular grid, average horizontal resolution ~3.5km
Vertical grid	21-layer terrain-following coordinate
Minimum depth	10m(ensure global stability)
Wave direction	Full-circle, 36 directional bands
Wave frequency	40 frequency bins, 0.04-1HZ
Time step	60s
Forcing	3 hourly NARR









1. Input

input	How
T, f	Set by researchers (settings)
H_i, C_ice	Calculated by CICE
U_10	Given by NARR

→

2. Run Exp.

Case	Method quantifying α	Modeling period	Note
EXP0	N/A	2014.01.01–2014.12.31	Wave-only modeling
EXP1	IC4M1	2014.01.01–2014.06.30	Coupled wave-ice modeling with both blocking effect and ice-induced wave attenuation
EXP2	IC4M2		
EXP3	IC4M3		
EXP4	IC4M4		
EXP5	IC4M5		
EXP6	IC4M6		
EXP7	IC4M7		
EXP8	N/A	2011.01.01–2011.03.03	Wave-only modeling
EXP9	IC4M6	2011.01.01–2011.03.03	Coupled wave-ice modeling with both blocking effect and ice-induced wave attenuation

IC4M6: Represents average of IC4M1-M7 method, Javaherian(2025)

→ SWH

※ Formula

Method	Formula
IC4M1	$\alpha = e^{(-0.18*T-0.73)}$
IC4M2	$\alpha = \frac{2.12*10^{-3}}{T^2} + \frac{4.59*10^{-2}}{T^1}$
IC4M3	$\alpha = e^{(-0.3203+2.058H_i-0.9375T-0.4269H_i^2+0.1566H_iT+0.0006T^2)}$
IC4M4	$\alpha = \begin{cases} 2*5.35*10^{-6} & (\mathrm{SWH}\leq 3) \\ \frac{2*16.05*10^{-6}}{\mathrm{SWH}} & (\mathrm{SWH} > 3) \end{cases}$
IC4M5	$\alpha = \begin{cases} 2*5.0*10^{-6} & (f < 0.1) \\ 2*7.0*10^{-6} & (0.1\leq f < 0.12) \\ 2*1.5*10^{-5} & (0.12\leq f < 0.16) \\ 2*1.0*10^{-4} & (f\geq 0.16) \end{cases}$
IC4M6 (IC4M6H)	$\alpha = \begin{cases} 2*2.94*10^{-6} & (f < 0.1) \\ 2*4.27*10^{-6} & (0.1\leq f < 0.15) \\ 2*7.95*10^{-6} & (0.15\leq f < 0.20) \\ 2*2.95*10^{-5} & (0.20\leq f < 0.25) \\ 2*1.12*10^{-4} & (0.25\leq f < 0.30) \\ 2*2.74*10^{-4} & (0.30\leq f < 0.35) \\ 2*4.95*10^{-4} & (0.35\leq f < 0.40) \\ 2*8.94*10^{-4} & (f\geq 0.40) \end{cases}$
IC4M7	$\alpha = 0.2 * T^{-2.13} * H_i$

→ $S_{ice} = -\alpha E C_g$

$$\frac{\partial N}{\partial t} + \nabla \cdot \left[\left(\vec{\mathbf{c}_g} + \vec{\mathbf{U}} \right) N \right] + \frac{\partial C_\sigma N}{\partial \sigma} + \frac{\partial C_\theta N}{\partial \theta} = \frac{(1-C_{\mathrm{ice}})\left(S_{\mathrm{in}} + S_{\mathrm{ds,w}} + S_{\mathrm{ds,br}}\right) + S_{\mathrm{nl}} + S_{\mathrm{ds,b}} + S_{\mathrm{ice}}}{\sigma},$$

3. Validation

Obs. source	Timeline	What is validated	Outputs of source
NDBC buoy (8 spots)	2014 May – Nov	SWAVE model (w/o ice) NARR model	SWH(= H_s), U_10
AWAC (Lake Erie 3 spots)	2011 Jan – Mar	CICE-SWAVE model	SWH(= H_s)
NIC satelite	Daily during ice seasons	CICE model(C_ice)	C_ice

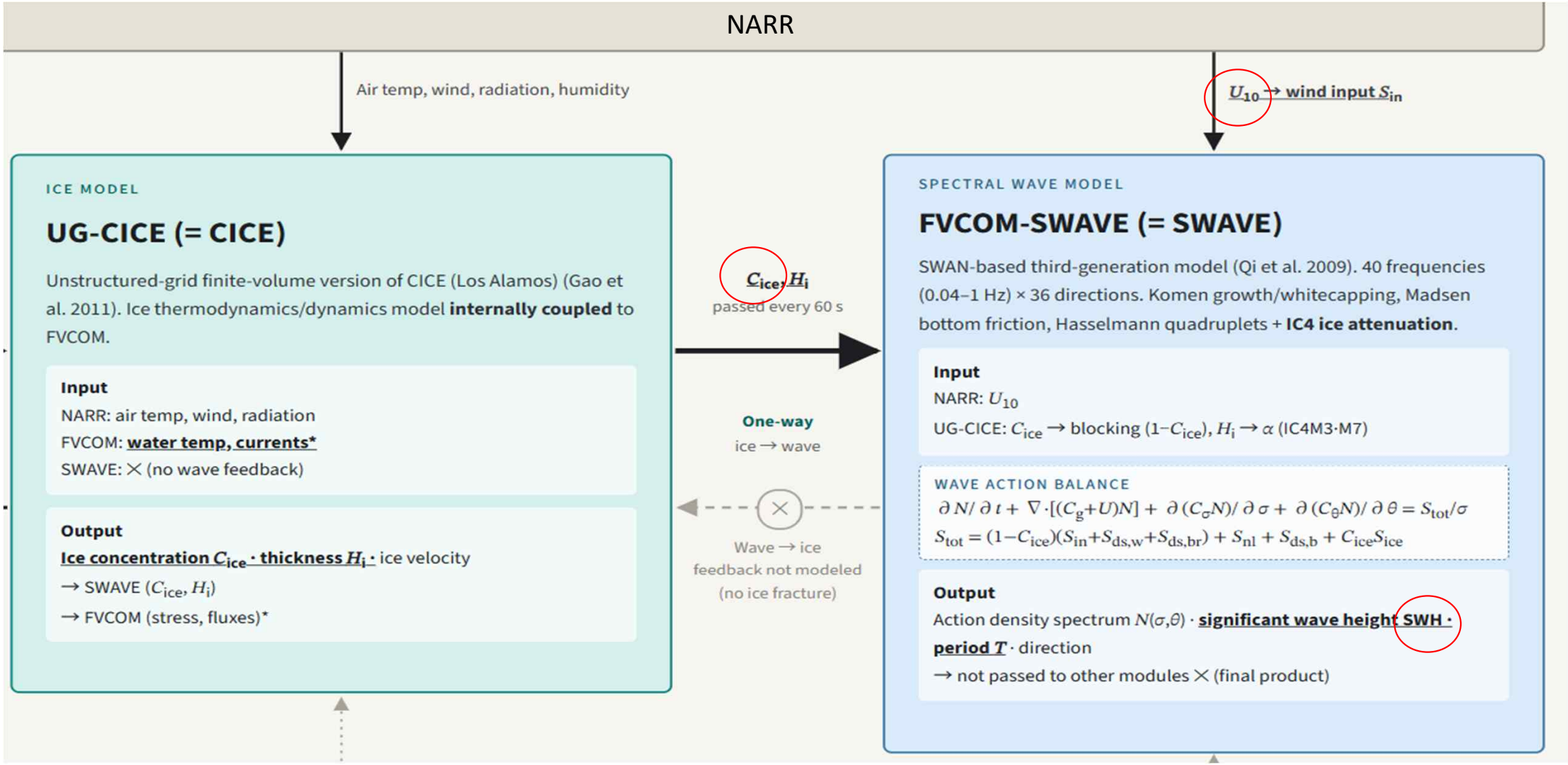
※ Evaluation Tools

$$\mathrm{CC} = \frac{\sum_{i=1}^{N}(x_i-x)(y_i-y)}{\sqrt{\frac{1}{N}\sum_{i=1}^{N}(x_i-x)^2}\sqrt{\frac{1}{N}\sum_{i=1}^{N}(y_i-y)^2}},$$

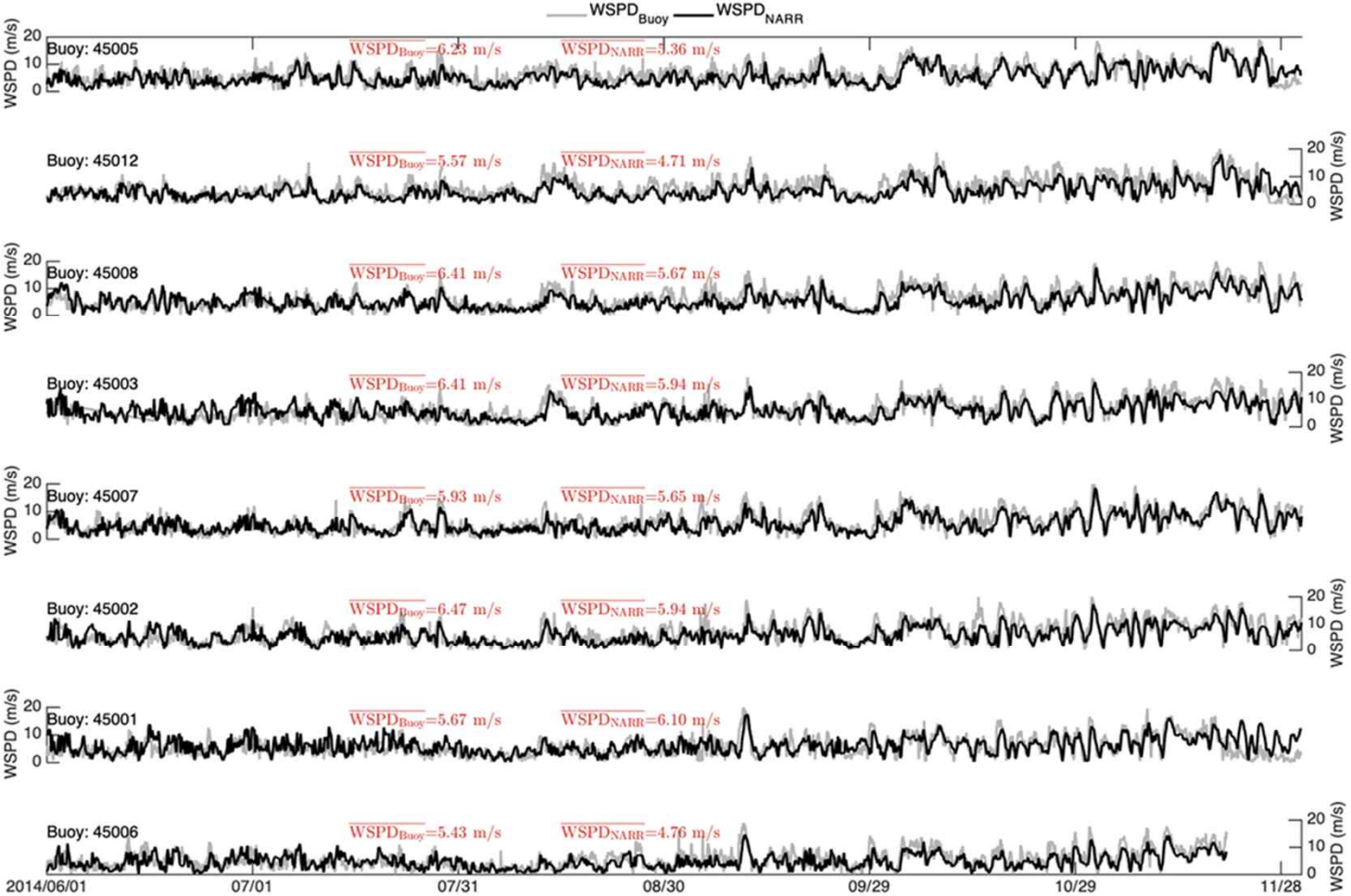
$$\mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(x_i-y_i)^2},$$

$$\mathrm{MAE} = \frac{1}{N}\sum_{i=1}^{N}|x_i-y_i|,$$

Numerical Experiments – Validation ※ Section 3



Numerical Experiments – Validation of Input (NARR wind speed)

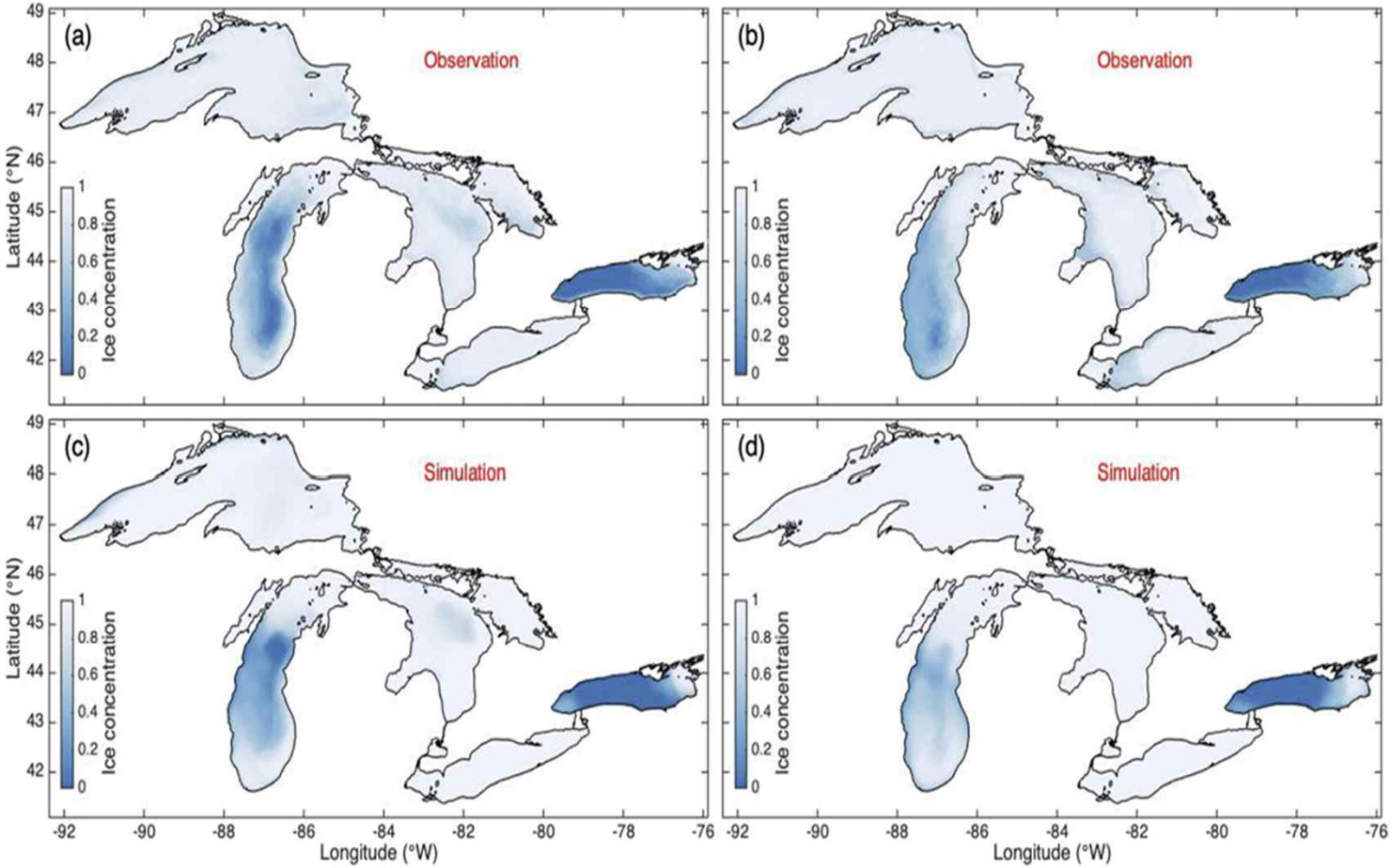


Obs. source	Timeline	What is validated	Outputs of source
NDBC buoy (8 spots)	2014 May – Nov	SWAVE model (w/o ice) NARR model	SWH(= H_s), U_10

NARR wind < observed wind

Fig. 4 Comparisons of the time series of wind speed between the NDBC buoy observations (gray lines) and the NARR product (black lines) during June 1 to December 1, 2014. The mean value of each wind speed time series is marked with red

Numerical Experiments – Validation of Input (C_ice)



Obs. source	Timeline	What is validated	Outputs of source
NIC satelite	Daily during ice seasons	CICE model(C_ice)	C_ice

Fig. 5 The monthly mean NIC satellite-measured ice concentration over the Great Lakes in February (**a**) and March (**b**), 2014, respectively. The simulated spatial distribution of monthly averaged ice concentration over the Great Lakes in February (**c**) and March (**d**), 2014, successively

Numerical Experiments – Validation of baseline [NDBC buoy data vs EXP0 (SWH)]

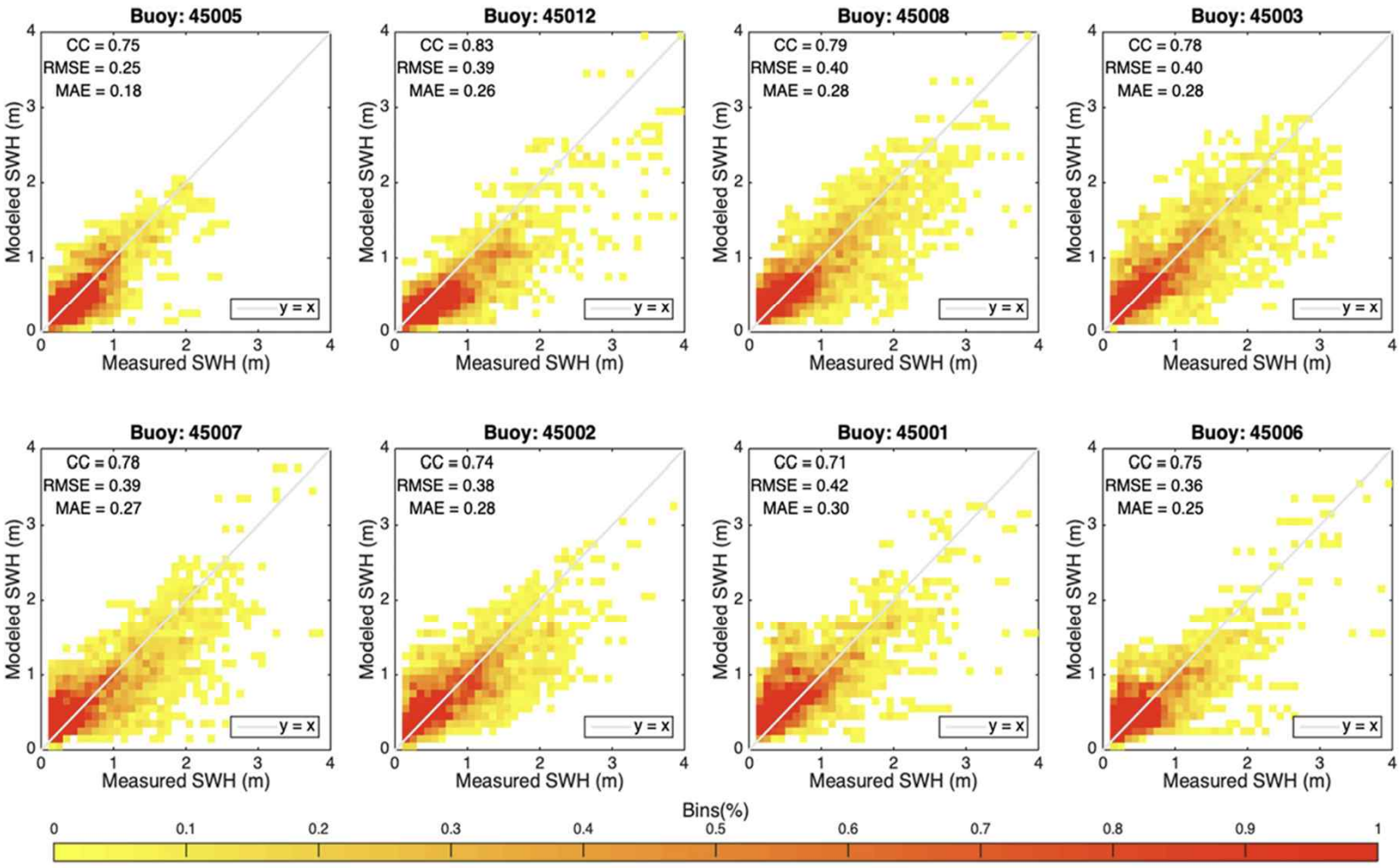
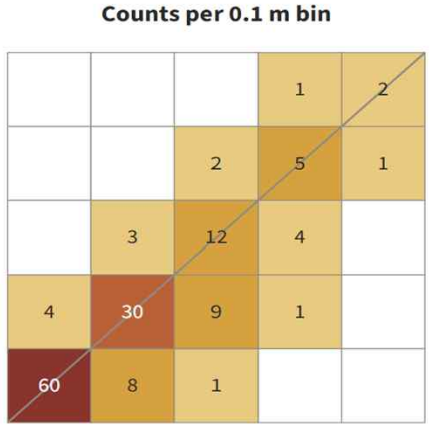


Fig. 3 Scatter diagrams of significant wave height: modeled results against the NDBC buoy observations. Scatter diagrams are created by binning the data into 0.1-m bins, and the gray line indicates function $y = x$. Model results are from numerical experiment EXP0 (a wave-only case)



Obs. source	Timeline	What is validated	Outputs of source
NDBC buoy (8 spots)	2014 May – Nov	SWAVE model (w/o ice) NARR model	SWH(= H_s), U_10

CC = 0.77
RMSE = 0.38m
MAE = 0.26m

Comparable with former Great Lakes wave simulations (Performance 'generally satisfactory')

Numerical Experiments – Comparison ※ Section 3

The structure is simple — 2 years × (control + treatment)

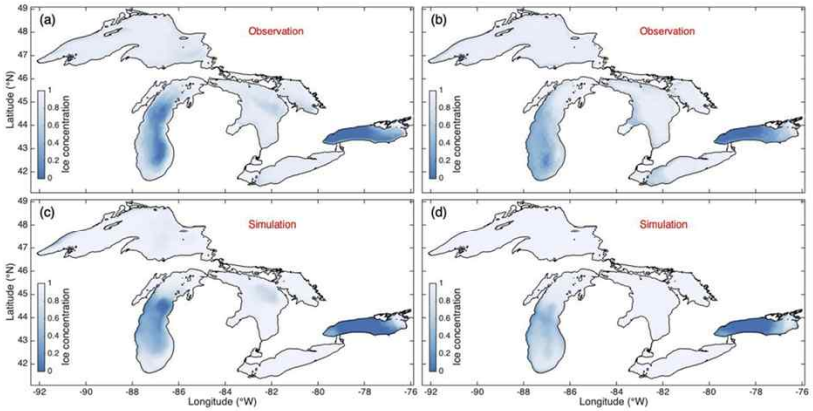
	2014	2011
Control (no ice physics)	**EXP0** wave-only	**EXP8** wave-only (same setup as EXP0)
Treatment (blocking + attenuation)	**EXP1–EXP7** = IC4M1–M7	**EXP9** = IC4M6 only

EXP8 is "identical" to EXP0 yet listed separately **because the year is different.**
It is the same setup re-run for 2011.

Role of the control — wave-only computes waves as if there were no ice at all. So

$$\text{EXP0} - \text{EXP1} = \text{the entire ice effect produced by IC4M1}$$

Numerical Experiments – Comparison of methods (SWH)



No observation data,
Only comparison between methods

EXP	Scheme	Average decrease
EXP3	IC4M3	0.467
EXP7	IC4M7	0.450
EXP2	IC4M2	0.446
EXP1	IC4M1	0.435
EXP6	IC4M6	0.424
EXP5	IC4M5	0.418
EXP4	IC4M4	0.324

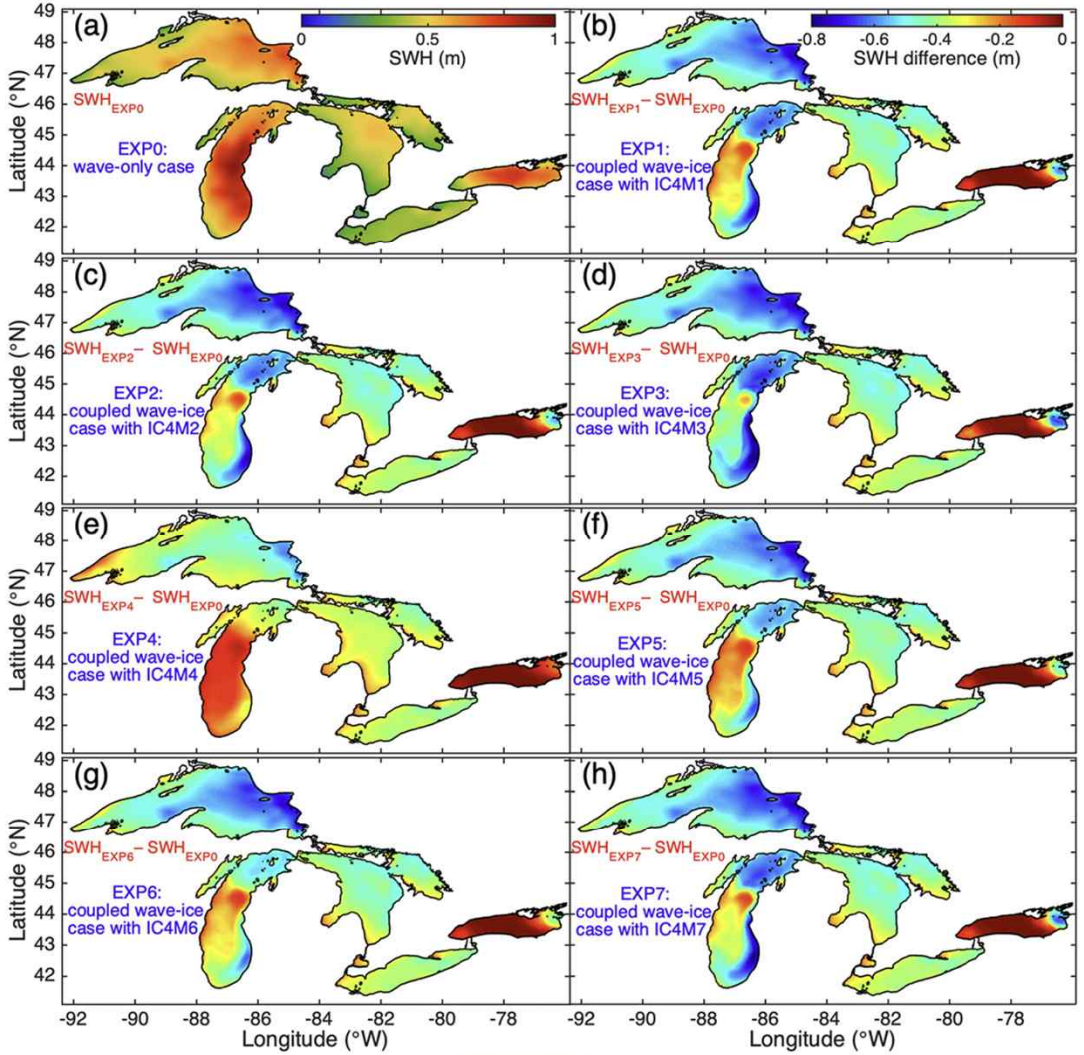
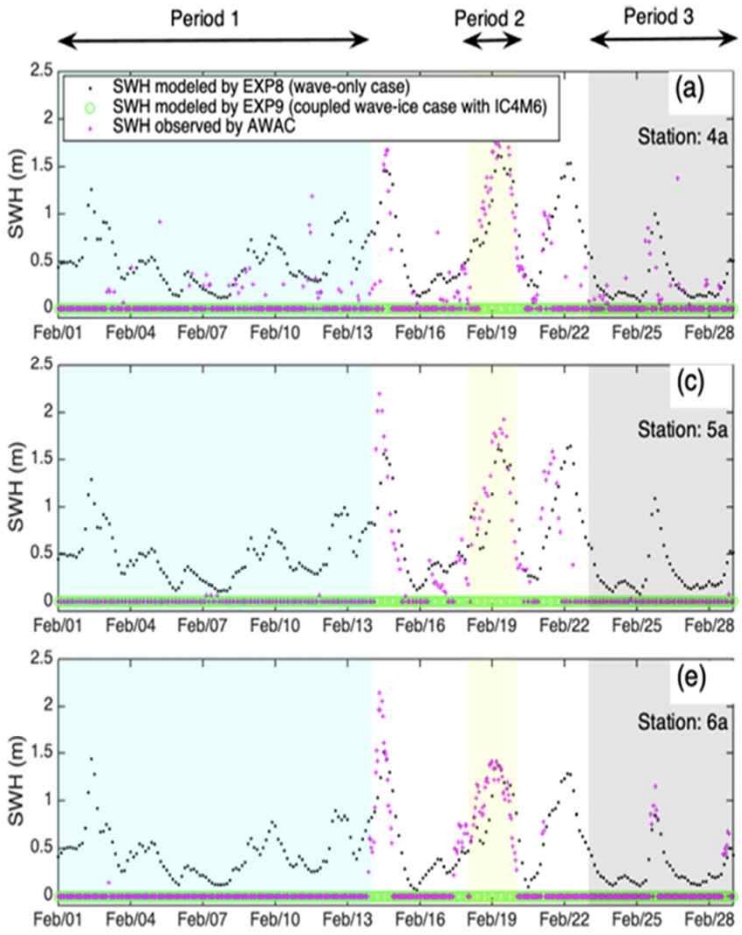


Fig. 6 Monthly mean SWH in the Great Lakes based on EXP0 for February 2014 (**a**). Monthly averaged differences in SWH between EXP0 and EXP1 (**b**), EXP2 (**c**), EXP3 (**d**), EXP4 (**e**), EXP5 (**f**), EXP6 (**g**), and EXP7 (**h**) in February 2014

Numerical Experiments – Comparison with 2011 Data (SWH)



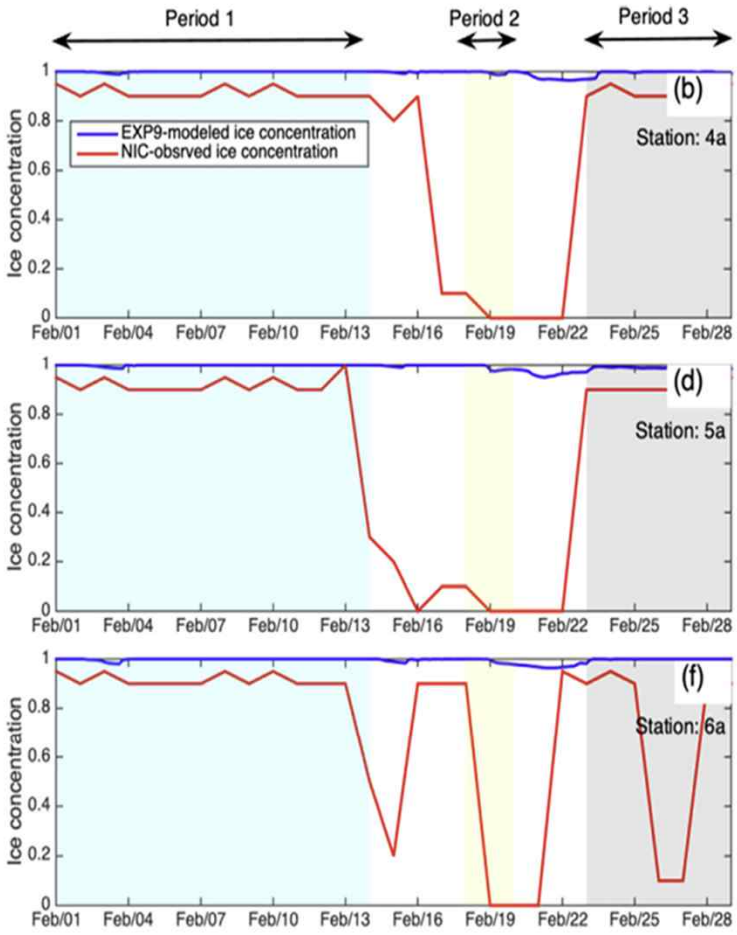
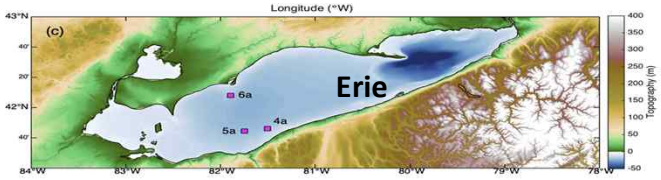


Fig. 7 Comparisons of the EXP8- and EXP9-modeled SWH with observations by AWAC at stations 4a (**a**), 5a (**c**), and 6a (**e**) in February 2011. The comparisons between the EXP9-modeled ice concentration and the NIC product at stations 4a (**b**), 5a (**d**), and 6a (**f**), respectively. Cyan, yellow, and gray shadings mark off Period 1 (February 01–February 14, 2011), Period 2 (February 18–February 20, 2011), and Period 3 (February 23–March 02, 2011), correspondingly



Obs. source	Timeline	What is validated	Outputs of source
AWAC (Lake Erie 3 spots)	2011 Jan – Mar	CICE-SWAVE model	SWH(= H_s)

		Period 1		Period 2		Period 3	
		RMSE (m)	MAE (m)	RMSE (m)	MAE (m)	RMSE (m)	MAE (m)
Station 4a	EXP8	0.54	0.47	*0.47*	*0.43*	0.34	0.27
	EXP9	*0.05*	*0.01*	1.32	1.11	*0.15*	*0.04*
Station 5a	EXP8	0.55	0.48	*0.40*	*0.36*	0.37	0.30
	EXP9	*0.01*	*0.00*	1.25	1.14	*0.01*	*0.00*
Station 6a	EXP8	0.52	0.46	*0.35*	*0.28*	0.27	0.23
	EXP9	*0.06*	*0.01*	1.02	0.95	*0.20*	*0.05*

Better statistical results are highlighted with italics

EXP9 fails.
Period2 shows limit of model.
Assumption of effect of wave → ice makes input C_ice wrong.

Limitations

- **C_ice error:** The wave → ice feedback was not considered (one-way coupling).

※ **Alternative validation:** What if NIC-observed C_ice had been used as input, and the resulting SWH compared with AWAC-observed SWH?

- **No validation of H_i:** Ice thickness was never validated.

- **Limitation of the experimental design:** Only EXP9 (IC4M6) was compared with AWAC data. If all of IC4M1–M7 (EXP1–7) had been run for 2011, the comparison would have shown which scheme best matches the observations.

Takeaways

- Obs. 2011 AWAC SWH data, same setting, same formula $S_{ice} = -\alpha E C_g$
- Inputs : C_ice, H_i, U_10
- Run : IC4M1-7 (or IC4M4) vs NN estimating α
- Output : SWH
- Compare: AWAC SWH vs generated SWH of SWAVE using estimated α by NN